Robots Need World Models That Know When the Future Forks: A Goal-Free Probe of Branching Fidelity

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Abstract: Some safety questions are counterfactual: would a slightly different execution of this action, or a slightly different version of the perceived scene, end in a different regime? Following the human intuitive physics engine [1], we answer them with a goal-free fork probe that samples the robot's measured execution and perception error, rolls out 64 futures, and alarms when they split into persistent, well-separated groups, with no task labels, failure data or fitted thresholds. In simulated Jenga extraction with real camera perception, the probe tracks ground-truth fragility on three fresh panels (κ 0.70–0.84 against a ceiling of 0.85–0.92, versus 0.32–0.51 when perception error is ignored), including when outcomes are read from image tokens. Used as an evaluation of world models, it shows that learned simulators accurate on average erase forks: a graph-network model halves fork recall, and a stochastic token-space world model reaches only κ 0.18 because its sampled spread is isotropic and ignores where futures branch. Robots need world models for such questions, but only ones with this branching fidelity, which current models lack and current benchmarks do not measure.

Keywords: World models, intuitive physics, runtime safety

1 Introduction

World models are usually judged by average prediction error. For safe manipulation, what often matters is whether a model knows where small, realistic deviations in action or perception lead to qualitatively different outcomes: the block slides out, or the tower tips. Such a fork is a narrow, high-consequence boundary, so a model can be accurate on average and still wrong exactly there. Asking whether a deviation would end in a different regime is a counterfactual query about futures the robot will not execute; an end-to-end policy does not answer it, a queryable model can.

Humans answer it with an intuitive physics engine (IPE) [1]: a few approximate, noisy simulations over an uncertain scene, whose disagreement signals precariousness; both perceptual and dynamics noise are needed to explain human judgments [2, 3]. We give a robot the same recipe and ask what it demands of perception and of the world model. Contributions: (1) a goal-free, label-free fork probe with a structure-only alarm; (2) an uncertainty-aware probe that samples the robot's measured perception error, confirmed with real camera perception on three fresh panels and working fully visually; (3) branching fidelity as a world-model evaluation, showing that learned simulators lose forks, with a diagnosis of why.

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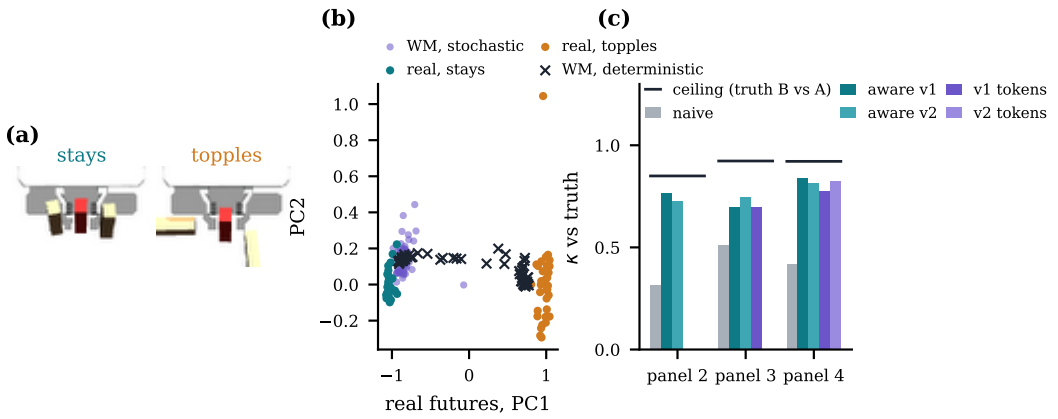


Figure 1: (a) Two of 64 rendered endings of one panel-3 state: the same action chunk under realistic execution error leaves the neighbour standing or topples it. (b) A held-out training fork (the most balanced real split): real latent futures (colour = outcome) on their two principal axes, with the token world models' futures for the same image and actions. The deterministic model spreads along the real branch axis; the stochastic model's samples stay near one outcome, with most of their spread outside this plane. (c) Agreement with fragility truth on three fresh panels; bars show the truth-vs-truth ceiling.

2 The fork probe

Setting and probe. A Franka Panda extracts a block from a Jenga-like stack in MuJoCo. At each chunk start the policy commits to an 8-step action chunk, followed by a 30-step hold. The probe draws 64 execution-error snippets from the robot's measured tracking residuals (median chunk maximum 3.2 mm), subtracts them from the commanded positions, and rolls out. It alarms if, at hold steps 10 and 30, the endings (object positions, or below, image tokens) projected on their first principal axis prefer two Gaussians to one by BIC, are well separated (Ashman's $D > 2$ [4]), have at least 2 rollouts in the minority, and keep the same partition. These are generic statistical conventions, not fitted to labels.

Uncertainty-aware probe (the IPE). The robot perceives $\hat{s}$, not the true scene s: a frozen DINOv2 encoder [5] on one close-camera frame plus a keypoint head regresses the poses of the three blocks in play (held-out error 2.5 mm). Its validation residuals define an error model (full 9×9 position covariance plus rotation covariances). The naive probe starts all rollouts at $\hat{s}$; the aware probe (v1) starts rollout j at $\hat{s} + e_j$ with e_j drawn from the error model; the Bayesian probe (v2) draws 5 hypotheses $\hat{s} + d_m$, runs an aware probe around each, and alarms if at least 3 split.

Ground truth. Fragility truth applies the aware recipe around the true state: a state is fragile if futures split within the robot's own uncertainty. Two independent seeds (A, B) agree at κ 0.85–0.92, the attainable ceiling. A harm fork is a state where 2–62 of 64 truth-A rollouts topple a neighbouring block. Each confirmation uses a fresh panel of new stack layouts (up to 60 topple forks plus 100 safe states) under a protocol frozen before data.

3 Results

The probe sees physical forks. With exact physics and the true state, the split test alarms on all 23 development topple forks at 10% alarms on quiet states.

Sampling the robot's own perception error restores calibrated judgments. Perception errors of 2–4 mm are comparable to execution noise, so the perceived scene is effectively a different scene and the naive probe misjudges fragility. Aware probes reach 76–91% of the ceiling, naive 38–55% (Table 1, Fig. 1c).

Table 1: Fork probes with real camera perception on three fresh panels (exact physics). κ against fragility truth A; harm = recall on harm forks; "tokens" = outcomes read from DINOv2 tokens of rendered endings (fully visual).

	Panel 2 (153 states)		Panel 3 (160)		Panel 4 (156)	
monitor	κ	harm	κ	harm	κ	harm
ceiling (truth B vs A)	0.85	–	0.92	–	0.92	–
naive ($\hat{s}$ only)	0.32	–	0.51	0.65	0.42	0.67
aware v1	0.76	–	0.70	0.77	0.84	0.91
aware v2 (Bayesian)	0.73	0.90	0.74	0.90	0.82	0.89
aware v1, tokens	–	–	0.70	0.77	0.77	0.88
aware v2, tokens	–	–	–	–	0.83	0.88

Table 2: Branching fidelity. Left: refined probe, true state, 84 held-out topple forks. Right: fresh panel 4 from images, and diagnostics on 60 held-out training states (spread relative to real; PC1 = share of spread on the main axis at real forks; ratio = spread on forks / non-forks).

dynamics	recall	quiet
exact physics	65.5%	3.5%
GNN (10 seeds)	31.1%	2.0%

dynamics (image input)	κ	harm	spread	PC1	ratio
real futures	–	–	1.00	0.57	4.8
MuJoCo + v2 tokens	0.83	0.88	–	–	–
token WM, stochastic	0.18	0.18	0.98	0.15	1.3
token WM, deterministic	0.42	0.60	0.04	0.69	32
[TODO: iteration 2]					

Reading outcomes from tokens costs little. The unchanged split test on flattened DINOv2 tokens of rendered endings agrees with the position-based test at κ 0.86–0.90 on aware rollouts; tokens ignore harmless sub-centimetre splits (22 of 24 disagreements on panel 2), acting as a consequence-scale floor. The fully visual Bayesian probe meets its frozen gate on panel 4 (harm recall 0.88, 7% alarms on non-harm states, κ 0.83). Misses come from where rollouts start, not from reading images: on panel 3, token- and position-read v1 each miss 14 of 62 harm forks, 12 of them because the single estimate put the whole rollout cloud on the safe side of the edge. Panels differ in difficulty, so v2's extra caution is not shown to be better calibrated than v1.

Branching fidelity of learned world models. Replacing the simulator by learned models, with everything else fixed, loses forks (Table 2). A graph-network state model trained with a counterfactual objective ranks fragile states well (AUC 0.97), yet under a refined probe keeps 31% of topple forks where physics keeps 65.5%. A stochastic token-space world model (DINOv2 tokens compressed by a fixed per-patch PCA to $28 \times 28 \times 8$, a flow-matching transition trained on 8 futures per image whose starts were sampled within perception error) fails on fresh panel 4 (κ 0.18, harm recall 0.18), below its own deterministic control (κ 0.42).

Why the stochastic model fails. It reproduces the amount of spread but not its shape or placement: the spread is isotropic and nearly the same on forks and non-forks, so it learned the average uncertainty of futures, not where they branch; other samplers are worse. The deterministic model's spread is tiny but aligned with the branch axis and 32× larger on forks (Fig. 1b); started from rendered perturbed scenes it doubles agreement with real futures (κ 0.17→0.37). As in the IPE, uncertainty should enter through initial conditions and be amplified by roughly right dynamics, not be added as per-step noise. A second iteration built on this (a learned operator perturbing the start latent by physical perception error, then deterministic dynamics trained on full rollouts) is under evaluation [TODO: result].

4 Related work

Simulation under uncertainty. Beyond the IPE and counterfactual accounts of physical support [6], robotics runs Monte Carlo rollouts over pose, friction or model parameters to score grasps and pushes [7, 8] or to build safety filters [9, 10], always against a task predicate or constraint. Zapolsky and Drumwright [11] proposed detecting divergent behaviour from particle traces but detect task failure in experiments. Our probe uses the robot's own measured residuals and outcome structure alone, a Monte Carlo test of proximity to a basin boundary [12]. Runtime monitors. Latent safety filters use learned failure classifiers, including on Jenga [13, 14]; failure detectors use policy uncertainty calibrated on successes [15, 16]. These measure what a model does not know; the fork probe measures whether the physics is at a branch. Physical fidelity of world models. Learned simulators are accurate on rigid contact [17, 18] and latent world models support planning [19, 20], but stiff contact is hard to learn [21] and audits find poor physical grounding and optimism under failure-inducing actions [22, 23]. We add a complementary axis.

5 Conclusion and limitations

A robot can judge fragility the way people do, by sampling its own perception and execution uncertainty through a rough simulator. For the question this workshop poses, our results suggest that pre-execution safety judgments need a queryable world model, but one with branching fidelity, which two learned families accurate on average lack. They also suggest where uncertainty should live in such a model: in the initial conditions, sampled from measured error, with deterministic and approximate dynamics. Average-error and success benchmarks would rank our failed models well; we propose branching fidelity, measured with the goal-free fork probe, as an additional evaluation axis. Limitations. Simulation only, one task family, three tracked blocks; the MuJoCo probes use privileged geometry except for the perceived poses; panels vary in difficulty; and the split test's statistical conventions act like fixed thresholds.

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